

More Advance SQL Queries

Learning Objectives:

- Understand Databases and SQLite through hands-on practice.

Instructor Expectations: During **DS5020**, it is essential to engage with me regularly by asking questions and sharing your progress in class. You will be required to share your screen so I can review your work and provide feedback. In-class work consists of two key components: the **Instructions** and the **Tasks to Do**. Please make sure to read them carefully.

Instructions:

1. **Attendance is MANDATORY** to ensure students gain hands-on experience with the material.
2. Assignments will be posted on Canvas under the "Assignments" section. Each assignment will include instructions, a due date, and a submission link for electronic submissions.
3. Please submit your work **before the deadline**. You're encouraged to upload your work early and often. Late submissions will be marked accordingly.
4. While working during class, you are required to **share your progress** with me. I will check it over and give you approval to continue.
5. I will provide feedback on what may be missing or need improvement. Make the necessary revisions before asking me for a final review and getting approval to submit your work.
6. Each assignment will test specific skills within the time provided in class. If you need more time to complete your work, please inform me via email.
7. Don't hesitate to **ask for help** at any time. This is your opportunity to learn, practice, and succeed.

Part 1: Simple Select Queries Across a Single Table

1. Retrieve all the university names.
2. What are the names and hiring dates of all faculty members?
3. Find the names of all departments in the database.
4. List the names of all courses offered.
5. How many seats are available in each course?

6. Retrieve the full names (first and last name) of all students.
7. List all students' emails and home addresses.

Export Data to CSV or Text File

You can use SQLite's .output command to export the results of a query into a file (CSV,Text)

Part 2: Select Queries Using Joins Between Two Tables

8. Which faculty members belong to each department?
9. List all the courses offered by each department.
10. Which students are enrolled in each course?
11. Retrieve the names of departments and the corresponding universities they belong to.
12. What are the courses and the faculty members teaching them?
13. Find the students and the departments they are enrolled in.
14. Which students are taking courses within the same department they are enrolled in?

Part 3: Select Queries Using Conditions, Aggregates, and Grouping

15. How many students are enrolled in each course?
16. List the total number of faculty members in each university.
17. Find the average number of credits per course.
18. Retrieve all the courses with more than 30 available seats.
19. Which universities have more than 5 faculty members?
20. How many courses are offered by each department?